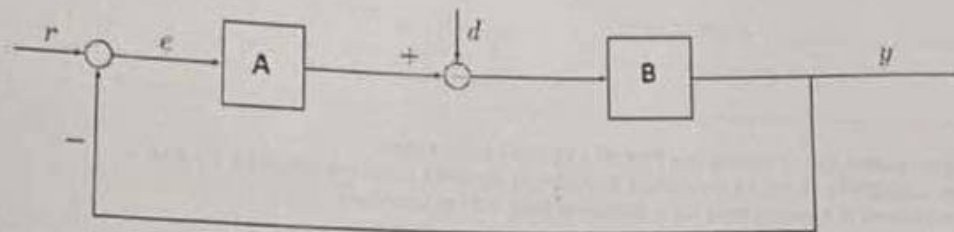


1	2	3	4	5	6	7	8	9	10

Q1) (20 points) Select the most correct answer from the alternatives given, and then write it in the table

1. You are given the Below system diagram. What is the associated transfer function from disturbance to output:

$$A = \frac{s+1}{s+2} \quad B = \ddot{y}(t) + 2\dot{y}(t) + y(t) = \dot{u}(t) + u(t)$$



a. $\frac{1}{s(s+1)}$

b. $\frac{s+1}{s+2}$

c. $\frac{s+1}{(s+2)(s+3)}$

d. $\frac{s+2}{(s+1)(s+3)}$

E. none

2. Which differential equation corresponds to the the following transfer function:

$$g(s) = \frac{y}{u} = \frac{s+2}{s^2+3s-4}$$

A. $\ddot{y} + 3\dot{y} - 4y = u + 2$

C. $\ddot{y} + 3\dot{y} - 4y = \dot{u} + 2u$

B. $-\ddot{y} - 3\dot{y} + 4y = u + 2$

D. $-\ddot{y} - 3\dot{y} + 4y = \dot{u} + 2u$

3. The transfer Function from Fig.1 is?

a. $A + BC + D$

b. $\frac{A+BC+D}{1+BC}$

c. $\frac{A+BC+D}{1+A+BC+D}$

d. $\frac{A+BC+D}{1+D}$

e. none

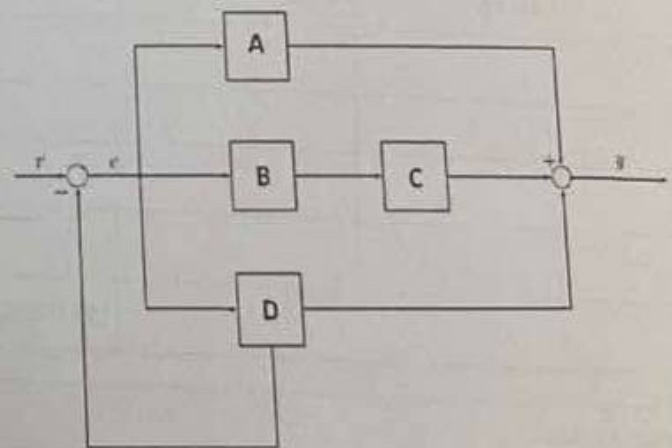


Fig 1

4. Which one of the following state space representations is equivalent to the system given by its transfer function $g(s) = \frac{s-1}{s^2+s-2}$?

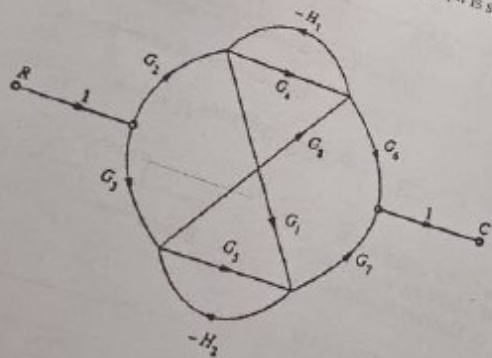
A. $A = \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix}, B = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, C = [1 \ 0], D = 1$

C. $A = \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix}, B = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, C = [1 \ 0], D = 0$

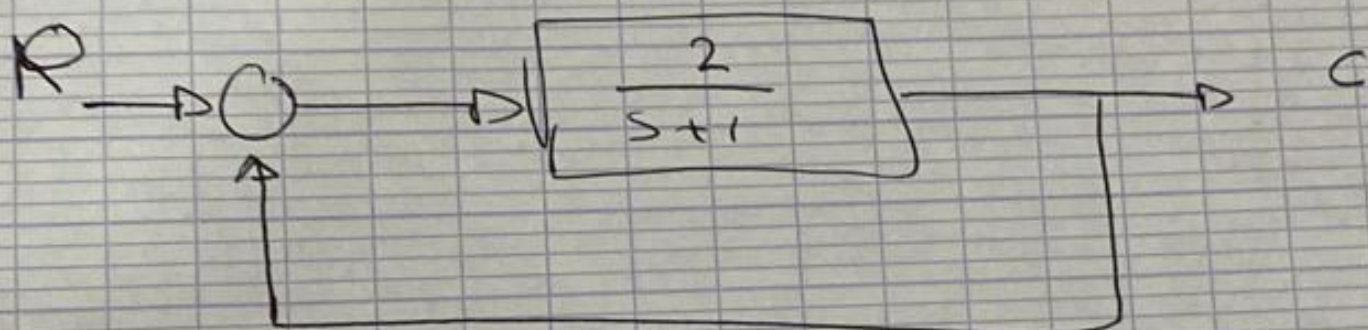
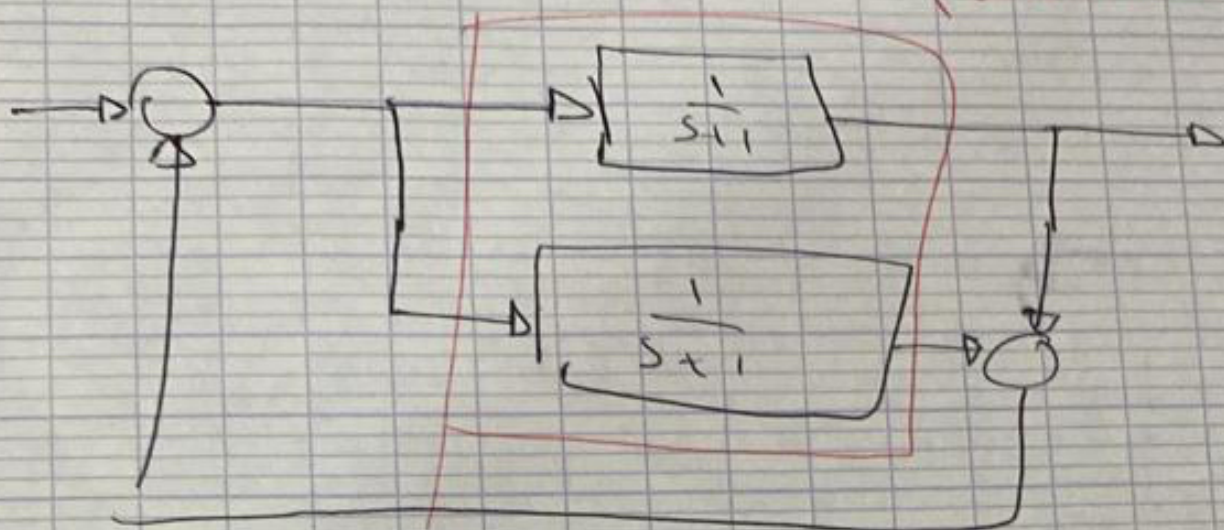
B. $A = \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix}, B = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, C = [0 \ 1], D = 0$

D. $A = \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix}, B = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, C = [0 \ 1], D = 1$

Q2) Consider the control system whose signal flow graph is shown below. Determine the system transfer function using Mason's formula. 10 Marks



Parallel



$$R \frac{C}{R} = \frac{\frac{2}{s+1}}{1 + \frac{2}{s+1}} \times \frac{\frac{2s+1}{2}}{\frac{s+1}{2}}$$

$$\frac{C}{R} = \frac{1}{\frac{s+1}{2} + 1} = \frac{1}{\frac{s+1+2}{2}}$$

$$\frac{C}{R} = \frac{2}{s+3}$$



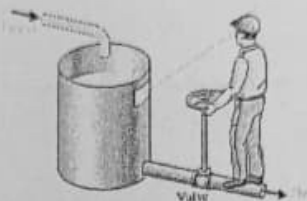
Question No. 1 (24 points)

Please select the most accurate answer for each of the following statements:

Q1

The control system below represents a

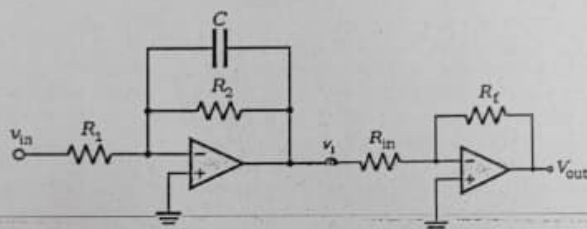
- a- Automatic Control b- Natural Control ☒ c- Manual Control d- None



Q2

For the operational amplifier circuit below, $V_{out}/V_{in} =$

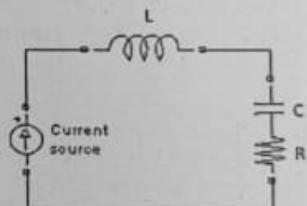
- ☒ a) $-R_f/R_{in}$ b) R_f/R_{in} c) $-R_f/(R_1 R_2 + C s + R_{in})$ d) None



Q3

For the circuit below, the equivalent impedance $Z_T =$

- a) $R + CS + LS$
b) $R + 1/CS$
☒ c) $R + LS + 1/CS$
d) None

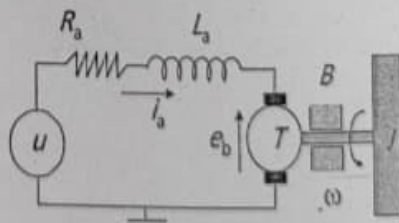


Q4

For the electromechanical system below, select the correct model:

(T_m : Motor torque, K_t : Torque Constant, K_b : Speed Constant, R_a : Armature Resistance, w : motor speed)

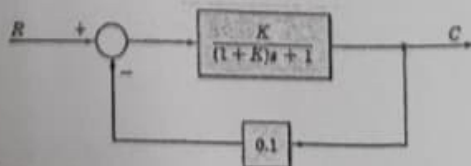
- a) $T_m = K_t i_a, L_a (di/dt) + R_a i_a + K_b w = U$
b) $T_m = K_t i_a, L_a (di/dt) + R_a i_a + K_b = U$
☒ c) $T_m = i_a, L_a (di/dt) + R_a i_a + K_b = U$
d) None



Q5

For the block diagram shown below $K=10$. The Transfer function is:

- a) $10/(11s+0.1)$
☒ b) $10/(11s+2)$
c) $0.1/(11s+1)$
d) None

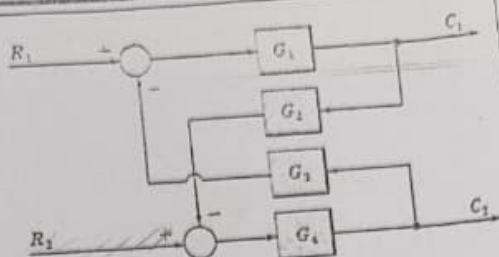




Q6

For the system below, the output C_1 due to the input R_1 is:

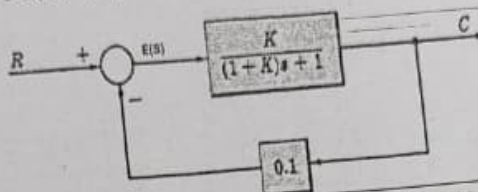
- a) $R_1 \frac{G_1}{1 - G_1 G_2 G_3 G_4}$
- b) $R_1 \frac{G_1}{1 - G_1 G_2 G_3 G_4}$
- c) $R_1 \frac{G_1}{1 + G_1 G_2 G_3 G_4}$
- d) None



Q7

For the block diagram shown below, $E(S)/R(S) =$

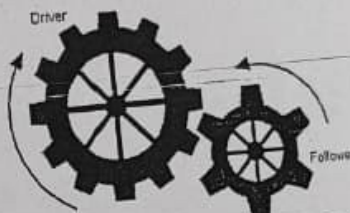
- a) $[(S+1)/((1+K)S+1+0.1K)]$
- b) $[(1+K)S+1]/[(1+K)S+1+0.1K]$
- c) $[(1+K)S+1]/[(K)S+1+0.1K]$
- d) None



Q8

For the gear system shown below, when the driver rotates 180° the follower rotates:

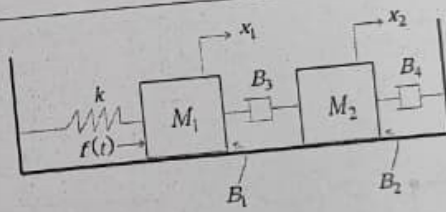
- a) 90°
- b) 45°
- c) 360°
- d) None



Q9

Select the equation that correctly describes the motion of the mass M_2

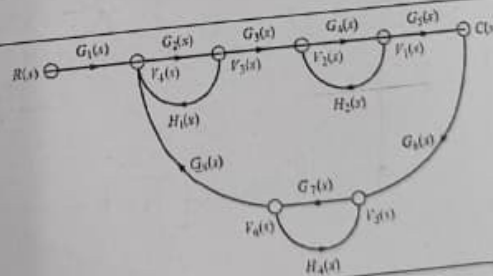
- a) $M_2 X_2'' = B_3(X_1 - X_2) - B_4 X_2' - B_2 X_2'$
- b) $M_2 X_2'' = B_3(X_1' - X_2') - B_4 X_2' - B_2 X_2'$
- c) $M_2 X_2'' = B_3(X_1 - X_2) - B_4 X_2'$
- d) None



10

The total number of loops (regardless of being touching or non-touching) in the signal flow graph below is:

- a) One loop
- b) Two loops
- c) Three loops
- d) Four loops



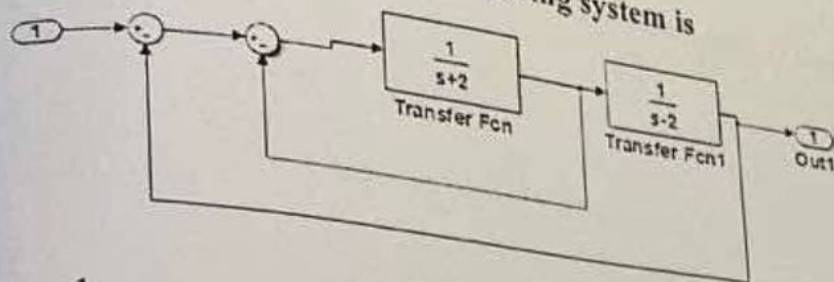
11

For the gear system shown below:

- a) The Driver and the Idler rotates in same direction.
- b) The Driver and the Follower rotates in same direction.
- c) The Idler and the Follower rotates in same direction.
- d) None



5. the transfer function of the following system is



A. $\frac{1}{s^2+s-6}$

B. $\frac{1}{s^2-s-6}$

C. $\frac{1}{s^2+s-5}$

D. $\frac{1}{s^2+s+6}$

E. none

6. moving the pole of the first order system near to the origin it will

A. make the system slower

B. Make the system faster

C. increase the Dc gain

D. Decreases the Dc gain

E. none

7. for the following first order system $\frac{C(s)}{R(s)} = \frac{10}{2s+1} e^{-2s}$ th Dc gain is

A. 5 B. 6.32 C. 0 D. 10 E. none

8. for the following first order system $\frac{C(s)}{R(s)} = \frac{10}{2s+1} e^{-2s}$ at $t=2$ the value of $c(t)$ is

A. 6.32 B. 2 C. 0 D. 4 E. none

9. A stable system is a dynamic system with a bounded output response for any input.

A. True

B. False

10. The shown block represent the following equation

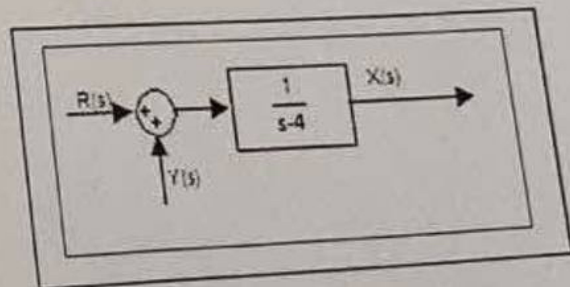
A. $r(t) = 4x(t) + x'(t) + y(t)$

B. $r(t) = 4x(t) + x'(t) - y'(t)$

C. $r(t) = x(t) - 4x'(t) + y(t)$

D. $r(t) = x(t) + 4x'(t) - y(t)$

E. none





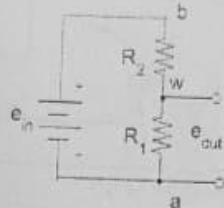
Q12

Following are equations representing the potentiometer output voltage e_{out} . Select the **WRONG** equation.

a) $e_{out} = e_{in} \frac{R_2}{R_1 + R_2}$

b) $e_{out} = e_{in} \frac{\theta}{\theta_{max}}$

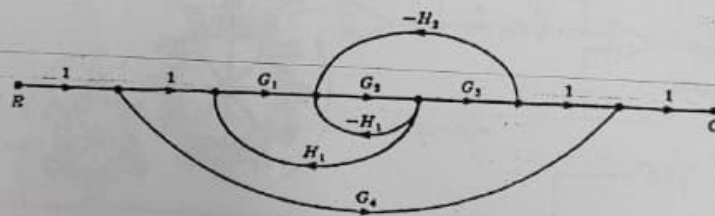
c) $e_{out} = e_{in} \frac{R_1}{R_1 + R_2}$



1	2	3	4	5	6	7	8	9	10	11	12
C	A	C	A	B	C	B	C	B	D	B	A

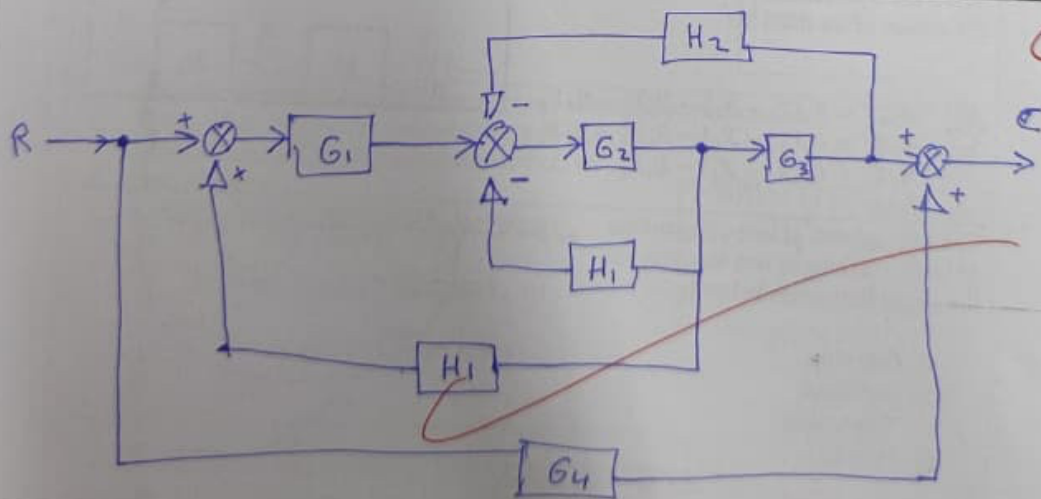
Q2 (6 points)

Represent the following Signal Flow graph as a Block Diagram and give the forward path.



22

Block Diagram



6

Forward Path

$G_1 G_2 G_3$